二维等参元高斯积分公式与例题：

几何模式：

$$x(\xi,\eta)=\sum_{i=1}^{n}N_i(\xi,\eta)x_i,\quad y(\xi,\eta)=\sum_{i=1}^{n}N_i(\xi,\eta)y_i$$

位移模式：

$$u(\xi,\eta)=\sum_{i=1}^{n}N_i(\xi,\eta)u_i,\quad v(\xi,\eta)=\sum_{i=1}^{n}N_i(\xi,\eta)v_i$$

$$\boldsymbol{J}=\begin{bmatrix}\dfrac{\partial x}{\partial\xi} & \dfrac{\partial y}{\partial\xi}\\ \dfrac{\partial x}{\partial\eta} & \dfrac{\partial y}{\partial\eta}\end{bmatrix}$$

$$|\boldsymbol{J}|=\frac{\partial x}{\partial\xi}\frac{\partial y}{\partial\eta}-\frac{\partial x}{\partial\eta}\frac{\partial y}{\partial\xi}$$

单元刚度矩阵：

$$[\boldsymbol{K}]^{\mathrm{e}}=\iint_{A}\left[\boldsymbol{B}(x,y)\right]^{\mathrm{T}}[\boldsymbol{D}]\left[\boldsymbol{B}(x,y)\right]t\,\mathrm{d}A$$

$$=\int_{-1}^{1}\int_{-1}^{1}\left[\boldsymbol{B}(\xi,\eta)\right]^{\mathrm{T}}[\boldsymbol{D}]\left[\boldsymbol{B}(\xi,\eta)\right]t\left|\boldsymbol{J}(\xi,\eta)\right|\mathrm{d}\xi\mathrm{d}\eta$$

$$=t\sum_{i=1}^{m}\sum_{j=1}^{n}w_iw_j\left[\boldsymbol{B}(\xi_i,\eta_j)\right]^{\mathrm{T}}[\boldsymbol{D}]\left[\boldsymbol{B}(\xi_i,\eta_j)\right]\left|\boldsymbol{J}(\xi_i,\eta_j)\right|$$

等效结点载荷（体力）

$$\{\boldsymbol{r}_b\}=t\iint_{A}\left[\boldsymbol{N}(x,y)\right]^{\mathrm{T}}\begin{Bmatrix}b_{\mathrm{x}}\\ b_{\mathrm{y}}\end{Bmatrix}\mathrm{d}A=$$

$$=t\int_{-1}^{1}\int_{-1}^{1}\left[\boldsymbol{N}(\xi,\eta)\right]^{\mathrm{T}}\begin{Bmatrix}b_{\mathrm{x}}\\ b_{\mathrm{y}}\end{Bmatrix}\left|\boldsymbol{J}(\xi,\eta)\right|\mathrm{d}\xi\mathrm{d}\eta$$

$$=t\sum_{i=1}^{m}\sum_{j=1}^{n}w_iw_j\left[\boldsymbol{N}(\xi_i,\eta_j)\right]^{\mathrm{T}}\begin{Bmatrix}b_{\mathrm{x}}\\ b_{\mathrm{y}}\end{Bmatrix}\left|\boldsymbol{J}(\xi_i,\eta_j)\right|$$

等效结点载荷（面力）

$$\{\boldsymbol{r}_q\} = t\int_c [\boldsymbol{N}_c]^{\mathrm{T}} \begin{Bmatrix} q_{\mathrm{x}} \\ q_{\mathrm{y}} \end{Bmatrix} \mathrm{d}c =$$

$$= t\int_{-1}^{1} [\boldsymbol{N}_c]^{\mathrm{T}} \begin{Bmatrix} q_{\mathrm{x}} \\ q_{\mathrm{y}} \end{Bmatrix} |J_c| \mathrm{d}a$$

$$= t\sum_{i=1}^{n} w_i \left[\boldsymbol{N}_c(a_i)\right]^{\mathrm{T}} \begin{Bmatrix} q_{\mathrm{x}} \\ q_{\mathrm{y}} \end{Bmatrix} |J_c(a_i)|$$

单元应变：

$$\{\boldsymbol{\varepsilon}\} = \begin{Bmatrix} \varepsilon_{\mathrm{x}} \\ \varepsilon_{\mathrm{y}} \\ \gamma_{\mathrm{xy}} \end{Bmatrix} = [\boldsymbol{B}(\xi,\eta)]\{\boldsymbol{\delta}\}^{\mathrm{e}}$$

单元应力：

$$\{\sigma\} = \begin{Bmatrix} \sigma_{\mathrm{x}} \\ \sigma_{\mathrm{y}} \\ \tau_{\mathrm{xy}} \end{Bmatrix} = [D][\boldsymbol{B}(\xi,\eta)]\{\boldsymbol{\delta}\}^{\mathrm{e}}$$

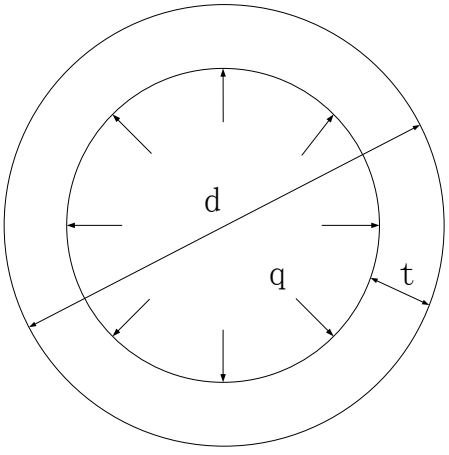
例题：

图示厚壁圆管，管长 L=2500 mm，管外径 d=100 mm，管厚 t=20 mm，圆管内壁受均布压力 q=1 Mpa，弹性模量 E=200 Gpa，泊松比 μ=0.3。试用等参单元求解：

（1） 结点位移。

（2） 单元高斯积分点位移（计算一个积分点即可）。

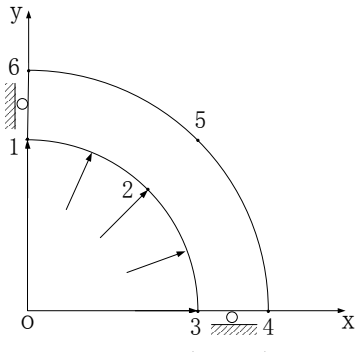
（3） 单元高斯积分点应力、主应力和 Von Mises 等效应力（计算一个积分点即可）。

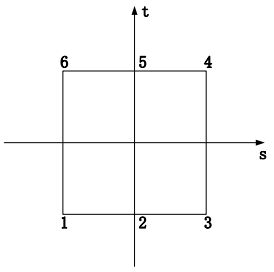


解：采用 mm-KN-GPa 单位制。

1）此问题为平面应变问题，取单位厚度管长进行分析即可。

又因为此结构为对称结构，取其 1/4 进行分析，如下图所示。





用一个六结点等参元进行分析，子单元结点号：1，2，3，4，5，6

结点坐标如下：

1：（0，30）； 2：$(15\sqrt{2},15\sqrt{2})$；3：（30，0）；

4：（50，0）； 5：$(25\sqrt{2},25\sqrt{2})$；6：（0，50）

插值函数为：

$$N_1(s,t)=-\frac{1}{4}(s-1)s(t-1);\ N_2(s,t)=\frac{1}{2}(s-1)(s+1)(t-1);\ N_3(s,t)=-\frac{1}{4}s(s+1)(t-1);$$

$$N_4(s,t)=\frac{1}{4}s(s+1)(t+1);\ N_5(s,t)=-\frac{1}{2}(s-1)(s+1)(t+1);\ N_6(s,t)=\frac{1}{4}(s-1)s(t+1)$$

单元几何模式：

$$x=\sum_{i=1}^{6}N_i x_i$$

$$=\frac{15\sqrt{2}}{2}(s-1)(s+1)(t-1)-\frac{15}{2}s(s+1)(t-1)+\frac{25}{2}s(s+1)(t+1)-\frac{25\sqrt{2}}{2}(s-1)(s+1)(t+1)$$

$$y=\sum_{i=1}^{6}N_i y_i$$

$$=-\frac{15}{2}(s-1)s(t-1)+\frac{15\sqrt{2}}{2}(s-1)(s+1)(t-1)-\frac{25\sqrt{2}}{2}(s-1)(s+1)(t+1)+\frac{25}{2}(s-1)s(t+1)$$

单元位移模式：

$$u=\sum_{i=1}^{6}N_i u_i \qquad\qquad v=\sum_{i=1}^{6}N_i v_i$$

雅可比矩阵：

$$[J]=\begin{bmatrix}\dfrac{\partial x}{\partial s} & \dfrac{\partial y}{\partial s}\\ \dfrac{\partial x}{\partial t} & \dfrac{\partial y}{\partial t}\end{bmatrix}=\begin{bmatrix}20.0+5.0t-16.6s-4.1st & -20.0-5.0t-16.6s-4.1st\\ 7.1+5.0s-2.1s^2 & 7.1-5.0s-2.1s^2\end{bmatrix}$$

$$|J|=282.8+70.7t+82.8s^2+20.7s^2t$$

$$[J]^{-1}=\frac{1}{|J|}\begin{bmatrix}7.1-5.0s-2.1s^2 & 20.0+5.0t+16.6s+4.1st\\ -7.1-5.0s+2.1s^2 & 20.0+5.0t-16.6s-4.1st\end{bmatrix}$$

应变矩阵：

$$[B]=\begin{bmatrix}1&0&0&0\\0&0&0&1\\0&1&1&0\end{bmatrix}\begin{bmatrix}[J]^{-1} & 0\\ 0 & [J]^{-1}\end{bmatrix}\begin{bmatrix}\dfrac{\partial N_1}{\partial s}&0&\dfrac{\partial N_2}{\partial s}&0&\dfrac{\partial N_3}{\partial s}&0&\dfrac{\partial N_4}{\partial s}&0&\dfrac{\partial N_5}{\partial s}&0&\dfrac{\partial N_6}{\partial s}&0\\ \dfrac{\partial N_1}{\partial t}&0&\dfrac{\partial N_2}{\partial t}&0&\dfrac{\partial N_3}{\partial t}&0&\dfrac{\partial N_4}{\partial t}&0&\dfrac{\partial N_5}{\partial t}&0&\dfrac{\partial N_6}{\partial t}&0\\ 0&\dfrac{\partial N_1}{\partial s}&0&\dfrac{\partial N_2}{\partial s}&0&\dfrac{\partial N_3}{\partial s}&0&\dfrac{\partial N_4}{\partial s}&0&\dfrac{\partial N_5}{\partial s}&0&\dfrac{\partial N_6}{\partial s}\\ 0&\dfrac{\partial N_1}{\partial t}&0&\dfrac{\partial N_2}{\partial t}&0&\dfrac{\partial N_3}{\partial t}&0&\dfrac{\partial N_4}{\partial t}&0&\dfrac{\partial N_5}{\partial t}&0&\dfrac{\partial N_6}{\partial t}\end{bmatrix}$$

其中：$\dfrac{\partial N_1}{\partial s}=-\dfrac{1}{4}(2s-1)(t-1)$，$\dfrac{\partial N_1}{\partial t}=-\dfrac{1}{4}(s-1)s$，$\dfrac{\partial N_2}{\partial s}=s(t-1)$，$\dfrac{\partial N_2}{\partial t}=\dfrac{1}{2}(s-1)(s+1)$，

$\dfrac{\partial N_3}{\partial s}=-\dfrac{1}{4}(2s+1)(t-1)$，$\dfrac{\partial N_3}{\partial t}=-\dfrac{1}{4}(s+1)s$，$\dfrac{\partial N_4}{\partial s}=\dfrac{1}{4}(2s+1)(t+1)$，$\dfrac{\partial N_4}{\partial t}=\dfrac{1}{4}(s+1)s$，

$\dfrac{\partial N_5}{\partial s}=-s(t+1)$，$\dfrac{\partial N_5}{\partial t}=-\dfrac{1}{2}(s-1)(s+1)$，$\dfrac{\partial N_6}{\partial s}=\dfrac{1}{4}(2s-1)(t+1)$，$\dfrac{\partial N_6}{\partial t}=\dfrac{1}{4}(s-1)s$。

弹性矩阵：

$$[D]=\frac{E}{(1+\mu)(1-2\mu)}\begin{bmatrix}1-\mu & \mu & 0\\ \mu & 1-\mu & 0\\ 0 & 0 & \dfrac{1-2\mu}{2}\end{bmatrix}$$

单元刚度矩阵：

$$[K]^e=\iint_A[B]^T[D][B]dA=\int_{-1}^{1}\int_{-1}^{1}[B]^T[D][B]|J|dsdt$$

记 $f(s,t)=[B]^T[D][B]|J|$

采用3×3高斯积分计算，有

$$[K]^e=\sum_{i=1}^{3}\sum_{j=1}^{3}w_iw_jf(s_i,t_j)$$

高斯积分点 1： $s_i=-0.774597,\ t_j=-0.774597,\ w_iw_j=0.308642$

$$[K]_1^e=\begin{bmatrix}
56.23 & 19.75 & -44.04 & 6.311 & 7.483 & -1.521 & 2.886 & 2.221 & -14.46 & -10.26 & -8.097 & -16.5 \\
19.75 & 29.56 & -1.982 & 3.064 & 0.05887 & -1.061 & 1.762 & 2.971 & -8.289 & -13.84 & -11.3 & -20.7 \\
-44.04 & -1.982 & 51.61 & -21.52 & -9.123 & 4.132 & -1.311 & -1.381 & 7.254 & 5.999 & -4.394 & 14.75 \\
6.311 & 3.064 & -21.52 & 32.88 & 3.999 & -6.315 & -0.5952 & 2.113 & 2.321 & -9.182 & 9.487 & -22.56 \\
7.483 & 0.0589 & -9.123 & 3.999 & 1.617 & -0.7628 & 0.2032 & 0.2273 & -1.149 & -0.9774 & 0.9679 & -2.545 \\
-1.521 & -1.061 & 4.132 & -6.315 & -0.7628 & 1.22 & 0.085 & -0.4506 & -0.3085 & 1.973 & -1.625 & 4.633 \\
2.886 & 1.762 & -1.311 & -0.5952 & 0.2032 & 0.085 & 0.2008 & 0.1339 & -0.9681 & -0.6397 & -1.011 & -0.7456 \\
2.221 & 2.971 & -1.381 & 2.113 & 0.2273 & -0.4506 & 0.1339 & 0.3987 & -0.6567 & -1.82 & -0.5449 & -3.212 \\
-14.46 & -8.289 & 7.254 & 2.321 & -1.149 & -0.3085 & -0.9681 & -0.6567 & 4.689 & 3.124 & 4.637 & 3.809 \\
-10.26 & -13.84 & 5.999 & -9.182 & -0.9774 & 1.973 & -0.6397 & -1.82 & 3.124 & 8.323 & 2.756 & 14.55 \\
-8.097 & -11.3 & -4.394 & 9.487 & 0.9679 & -1.625 & -1.011 & -0.5449 & 4.637 & 2.756 & 7.897 & 1.229 \\
-16.5 & -20.7 & 14.75 & -22.56 & -2.545 & 4.633 & -0.7456 & -3.212 & 3.809 & 14.55 & 1.229 & 27.29
\end{bmatrix}$$

高斯积分点 2： $s_i=-0.774597,\ t_j=0,\ w_iw_j=0.493827$

$$[K]_2^e=\begin{bmatrix}
41.39 & 24.78 & -11.75 & 4.999 & 1.572 & -1.352 & 7.02 & 3.611 & -36.72 & -17.74 & -1.507 & -14.3 \\
24.78 & 48.82 & -2.478 & 26.42 & 0.07236 & -5.822 & 3.979 & 6.298 & -20.38 & -29.11 & -5.976 & -46.61 \\
-11.75 & -2.478 & 22.25 & -14.04 & -4.049 & 2.714 & -2.924 & 0.3047 & 17.09 & -2.996 & -20.61 & 16.49 \\
4.999 & 26.42 & -14.04 & 30.34 & 2.593 & -6.206 & 1.468 & 2.621 & -8.881 & -10.11 & 13.86 & -43.07 \\
1.572 & 0.07236 & -4.049 & 2.593 & 0.7461 & -0.4953 & 0.4438 & -0.1163 & -2.664 & 0.857 & 3.951 & -2.911 \\
-1.352 & -5.822 & 2.714 & -6.206 & -0.4953 & 1.276 & -0.344 & -0.6009 & 2.021 & 2.395 & -2.543 & 8.957 \\
7.02 & 3.979 & -2.924 & 1.468 & 0.4438 & -0.344 & 1.237 & 0.5471 & -6.556 & -2.614 & 0.7793 & -3.036 \\
3.611 & 6.298 & 0.3047 & 2.621 & -0.1163 & -0.6009 & 0.5471 & 0.8514 & -2.735 & -4.034 & -1.612 & -5.135 \\
-36.72 & -20.38 & 17.09 & -8.881 & -2.664 & 2.021 & -6.556 & -2.735 & 34.93 & 12.91 & -6.077 & 17.06 \\
-17.74 & -29.11 & -2.996 & -10.11 & 0.857 & 2.395 & -2.614 & -4.034 & 12.91 & 19.35 & 9.586 & 21.51 \\
-1.507 & -5.976 & -20.61 & 13.86 & 3.951 & -2.543 & 0.7793 & -1.612 & -6.077 & 9.586 & 23.47 & -13.31 \\
-14.3 & -46.61 & 16.49 & -43.07 & -2.911 & 8.957 & -3.036 & -5.135 & 17.06 & 21.51 & -13.31 & 64.35
\end{bmatrix}$$

高斯积分点 3： $s_i=-0.774597,\ t_j=0.774597,\ w_iw_j=0.308642$

$$[K]_3^e = \begin{bmatrix}
14.33 & 8.544 & 6.467 & 3.193 & -1.463 & -0.7462 & 3.456 & 2.639 & -17.71 & -13.87 & -5.082 & 0.24 \\
8.544 & 37.53 & 2.14 & 22.59 & -0.5456 & -4.909 & 3.558 & 4.117 & -19.14 & -18.12 & 5.44 & -41.2 \\
6.467 & 2.14 & 4.002 & 0.02069 & -0.8668 & -0.03847 & 0.6135 & 1.341 & -2.573 & -7.369 & -7.643 & 3.906 \\
3.193 & 22.59 & 0.02069 & 14.01 & -0.05546 & -3.033 & 2.01 & 2.116 & -11.05 & -8.835 & 5.879 & 26.84 \\
-1.463 & -0.5456 & -0.8668 & -0.05546 & 0.1888 & 0.01937 & -0.1727 & -0.2981 & 0.776 & 1.628 & 1.538 & -0.7481 \\
-0.7462 & -4.909 & -0.03847 & -3.033 & 0.01937 & 0.6571 & -0.4403 & -0.4696 & 2.413 & 1.976 & -1.208 & 5.779 \\
3.456 & 3.558 & 0.6135 & 2.01 & -0.1727 & -0.4403 & 1.659 & 0.5051 & -9.001 & -2.376 & 3.445 & -3.257 \\
2.639 & 4.117 & 1.341 & 2.116 & -0.2981 & -0.4696 & 0.5051 & 0.767 & -2.509 & -3.794 & -1.677 & -2.736 \\
-17.71 & -19.14 & -2.573 & -11.05 & 0.776 & 2.413 & -9.001 & -2.509 & 48.98 & 11.59 & -20.47 & 18.69 \\
-13.87 & -18.12 & -7.369 & -8.835 & 1.628 & 1.976 & -2.376 & -3.794 & 11.59 & 19.09 & 10.4 & 9.677 \\
-5.082 & 5.44 & -7.643 & 5.879 & 1.538 & -1.208 & 3.445 & -1.677 & -20.47 & 10.4 & 28.21 & -18.83 \\
0.24 & -41.2 & 3.906 & -26.84 & -0.7481 & 5.779 & -3.257 & -2.736 & 18.69 & 9.677 & -18.83 & 55.33
\end{bmatrix}$$

高斯积分点 4： $s_i = 0,\ t_j = -0.774597,\ w_i w_j = 0.493827$

$$[K]_4^e = \begin{bmatrix}
7.376 & -4.098 & 10.53 & 2.107 & -7.376 & 4.098 & -0.9369 & 0.5205 & -10.53 & -2.107 & 0.9369 & -0.5205 \\
-4.098 & 7.376 & -2.107 & -10.53 & 4.098 & -7.376 & 0.5205 & -0.9369 & 2.107 & 10.53 & -0.5205 & 0.9369 \\
10.53 & -2.107 & 48.73 & 27.07 & -10.53 & 2.107 & -1.338 & 0.2676 & -48.73 & -27.07 & 1.338 & -0.2676 \\
2.107 & -10.53 & 27.07 & 48.73 & -2.107 & 10.53 & -0.2676 & 1.338 & -27.07 & -48.73 & 0.2676 & -1.338 \\
-7.376 & 4.098 & -10.53 & -2.107 & 7.376 & -4.098 & 0.9369 & -0.5205 & 10.53 & 2.107 & -0.9369 & 0.5205 \\
4.098 & -7.376 & 2.107 & 10.53 & -4.098 & 7.376 & -0.5205 & 0.9369 & -2.107 & -10.53 & 0.5205 & -0.9369 \\
-0.9369 & 0.5205 & -1.338 & -0.2676 & 0.9369 & -0.5205 & 0.119 & -0.06611 & 1.338 & 0.2676 & -0.119 & 0.06611 \\
0.5205 & -0.9369 & 0.2676 & 1.338 & -0.5205 & 0.9369 & -0.06611 & 0.119 & -0.2676 & -1.338 & 0.06611 & -0.119 \\
-10.53 & 2.107 & -48.73 & -27.07 & 10.53 & -2.107 & 1.338 & -0.2676 & 48.73 & 27.07 & -1.338 & 0.2676 \\
-2.107 & 10.53 & -27.07 & -48.73 & 2.107 & -10.53 & 0.2676 & -1.338 & 27.07 & 48.73 & -0.2676 & 1.338 \\
0.9369 & -0.5205 & 1.338 & 0.2676 & -0.9369 & 0.5205 & -0.119 & 0.06611 & -1.338 & -0.2676 & 0.119 & -0.06611 \\
-0.5205 & 0.9369 & -0.2676 & -1.338 & 0.5205 & -0.9369 & 0.06611 & -0.119 & 0.2676 & 1.338 & -0.06611 & 0.119
\end{bmatrix}$$

高斯积分点 5： $s_i = 0,\ t_j = 0,\ w_i w_j = 0.790123$

$$[K]_5^e = \begin{bmatrix}
3.022 & -1.679 & 9.497 & 1.899 & -3.022 & 1.679 & -3.022 & 1.679 & -9.497 & -1.899 & 3.022 & -1.679 \\
-1.679 & 3.022 & -1.899 & -9.497 & 1.679 & -3.022 & 1.679 & -3.022 & 1.899 & 9.497 & -1.679 & 3.022 \\
9.497 & -1.899 & 96.7 & 53.72 & -9.497 & 1.899 & -9.497 & 1.899 & -96.7 & -53.72 & 9.497 & -1.899 \\
1.899 & -9.497 & 53.72 & 96.7 & -1.899 & 9.497 & -1.899 & 9.497 & -53.72 & -96.7 & 1.899 & -9.497 \\
-3.022 & 1.679 & -9.497 & -1.899 & 3.022 & -1.679 & 3.022 & -1.679 & 9.497 & 1.899 & -3.022 & 1.679 \\
1.679 & -3.022 & 1.899 & 9.497 & -1.679 & 3.022 & -1.679 & 3.022 & -1.899 & -9.497 & 1.679 & -3.022 \\
-3.022 & 1.679 & -9.497 & -1.899 & 3.022 & -1.679 & 3.022 & -1.679 & 9.497 & 1.899 & -3.022 & 1.679 \\
1.679 & -3.022 & 1.899 & 9.497 & -1.679 & 3.022 & -1.679 & 3.022 & -1.899 & -9.497 & 1.679 & -3.022 \\
-9.497 & 1.899 & -96.7 & -53.72 & 9.497 & -1.899 & 9.497 & -1.899 & 96.7 & 53.72 & -9.497 & 1.899 \\
-1.899 & 9.497 & -53.72 & -96.7 & 1.899 & -9.497 & 1.899 & -9.497 & 53.72 & 96.7 & - 1.899 & 9.497 \\
3.022 & -1.679 & 9.497 & 1.899 & -3.022 & 1.679 & -3.022 & 1.679 & -9.497 & -1.899 & 3.022 & -1.679 \\
-1.679 & 3.022 & -1.899 & -9.497 & 1.679 & -3.022 & 1.679 & -3.022 & 1.899 & 9.497 & -1.679 & 3.022
\end{bmatrix}$$

高斯积分点 6： $s_i = 0,\ t_j = 0.774597,\ w_i w_j = 0.493827$

$$[K]_6^e = \begin{bmatrix}
0.08039 & -0.04466 & 1.338 & 0.2676 & -0.08039 & 0.04466 & -0.6329 & 0.3516 & -1.338 & -0.2676 & 0.6329 & -0.3516 \\
-0.04466 & 0.08039 & -0.2676 & -1.338 & 0.04466 & -0.08039 & 0.3516 & -0.6329 & 0.2676 & 1.338 & -0.3516 & 0.6329 \\
1.338 & -0.2676 & 72.14 & 40.08 & -1.338 & 0.2676 & -10.53 & 2.107 & -72.14 & -40.08 & 10.53 & -2.107 \\
0.2676 & -1.338 & 40.08 & 72.14 & -0.2676 & 1.338 & -2.107 & 10.53 & -40.08 & -72.14 & 2.107 & -10.53 \\
-0.08039 & 0.04466 & -1.338 & -0.2676 & 0.08039 & -0.04466 & 0.6329 & -0.3516 & 1.338 & 0.2676 & -0.6329 & 0.3516 \\
0.04466 & -0.08039 & 0.2676 & 1.338 & -0.04466 & 0.08039 & -0.3516 & 0.6329 & -0.2676 & -1.338 & 0.3516 & -0.6329 \\
-0.6329 & 0.3516 & -10.53 & -2.107 & 0.6329 & -0.3516 & 4.983 & -2.768 & 10.53 & 2.107 & -4.983 & 2.768 \\
0.3516 & -0.6329 & 2.107 & 10.53 & -0.3516 & 0.6329 & -2.768 & 4.983 & -2.107 & -10.53 & 2.768 & -4.983 \\
-1.338 & 0.2676 & -72.14 & -40.08 & 1.338 & -0.2676 & 10.53 & -2.107 & 72.14 & 40.08 & -10.53 & 2.107 \\
-0.2676 & 1.338 & -40.08 & -72.14 & 0.2676 & -1.338 & 2.107 & -10.53 & 40.08 & 72.14 & -2.107 & 10.53 \\
0.6329 & -0.3516 & 10.53 & 2.107 & -0.6329 & 0.3516 & -4.983 & 2.768 & -10.53 & -2.107 & 4.983 & -2.768 \\
-0.3516 & 0.6329 & -2.107 & -10.53 & 0.3516 & -0.6329 & 2.768 & -4.983 & 2.107 & 10.53 & -2.768 & 4.983
\end{bmatrix}$$

高斯积分点 7： $s_i = 0.774597,\ t_j = -0.774597,\ w_i w_j = 0.308642$

$$[K]_7^e = \begin{bmatrix}
1.22 & -0.7628 & -6.315 & 4.132 & -1.061 & -1.521 & 4.633 & -1.625 & 1.973 & -0.3085 & -0.4506 & 0.08499 \\
-0.7628 & 1.617 & 3.999 & -9.123 & 0.05887 & 7.483 & -2.545 & 0.9679 & -0.9774 & -1.149 & 0.2273 & 0.2032 \\
-6.315 & 3.999 & 32.88 & -21.52 & 3.064 & 6.311 & -22.56 & 9.487 & -9.182 & 2.321 & 2.113 & -0.5952 \\
4.132 & -9.123 & -21.52 & 51.61 & -1.982 & -44.04 & 14.75 & -4.394 & 5.999 & 7.254 & -1.381 & -1.311 \\
-1.061 & 0.05887 & 3.064 & -1.982 & 29.56 & 19.75 & -20.7 & -11.3 & -13.84 & -8.289 & 2.971 & 1.762 \\
-1.521 & 7.483 & 6.311 & -44.04 & 19.75 & 56.23 & -16.5 & -8.097 & -10.26 & -14.46 & 2.221 & 2.886 \\
4.633 & -2.545 & -22.56 & 14.75 & -20.7 & -16.5 & 27.29 & 1.229 & 14.55 & 3.809 & -3.212 & -0.7456 \\
-1.625 & 0.9679 & 9.487 & -4.394 & -11.3 & -8.097 & 1.229 & 7.897 & 2.756 & 4.637 & -0.5449 & -1.011 \\
1.973 & -0.9774 & -9.182 & 5.999 & -13.84 & -10.26 & 14.55 & 2.756 & 8.323 & 3.124 & -1.82 & -0.6397 \\
-0.3085 & -1.149 & 2.321 & 7.254 & -8.289 & -14.46 & 3.809 & 4.637 & 3.124 & 4.689 & -0.6567 & -0.9681 \\
-0.4506 & 0.2273 & 2.113 & -1.381 & 2.971 & 2.221 & -3.212 & -0.5449 & -1.82 & -0.6567 & 0.3987 & 0.1339 \\
0.08499 & 0.2032 & -0.5952 & -1.311 & 1.762 & 2.886 & -0.7456 & -1.011 & -0.6397 & -0.9681 & 0.1339 & 0.2008
\end{bmatrix}$$

高斯积分点 8： $s_i = 0.774597,\ t_j = 0,\ w_i w_j = 0.493827$

$$[K]_8^e = \begin{bmatrix}
1.276 & -0.4953 & -6.206 & 2.714 & -5.822 & -1.352 & 8.957 & -2.543 & 2.395 & 2.021 & -0.6009 & -0.344 \\
-0.4953 & 0.7461 & 2.593 & -4.049 & 0.07236 & 1.572 & -2.911 & 3.951 & 0.857 & -2.664 & -0.1163 & 0.4438 \\
-6.206 & 2.593 & 30.34 & -14.04 & 26.42 & 4.999 & -43.07 & 13.86 & -10.11 & -8.881 & 2.621 & 1.468 \\
2.714 & -4.049 & -14.04 & 22.25 & -2.478 & -11.75 & 16.49 & -20.61 & -2.996 & 17.09 & 0.3047 & -2.924 \\
-5.822 & 0.07236 & 26.42 & -2.478 & 48.82 & 24.78 & -46.61 & -5.976 & -29.11 & -20.38 & 6.298 & 3.979 \\
-1.352 & 1.572 & 4.999 & -11.75 & 24.78 & 41.39 & -14.3 & -1.507 & -17.74 & -36.72 & 3.611 & 7.02 \\
8.957 & -2.911 & -43.07 & 16.49 & -46.61 & -14.3 & 64.35 & -13.31 & 21.51 & 17.06 & -5.135 & -3.036 \\
-2.543 & 3.951 & 13.86 & -20.61 & -5.976 & -1.507 & -13.31 & 23.47 & 9.586 & -6.077 & -1.612 & 0.7793 \\
2.395 & 0.857 & -10.11 & -2.996 & -29.11 & -17.74 & 21.51 & 9.586 & 19.35 & 12.91 & -4.034 & -2.614 \\
2.021 & -2.664 & -8.881 & 17.09 & -20.38 & -36.72 & 17.06 & -6.077 & 12.91 & 34.93 & -2.735 & -6.556 \\
-0.6009 & -0.1163 & 2.621 & 0.3047 & 6.298 & 3.611 & -5.135 & -1.612 & -4.034 & -2.735 & 0.8514 & 0.5471 \\
-0.344 & 0.4438 & 1.468 & -2.924 & 3.979 & 7.02 & -3.036 & 0.7793 & -2.614 & -6.556 & 0.5471 & 1.237
\end{bmatrix}$$

高斯积分点 9： $s_i = 0.774597,\ t_j = 0.774597,\ w_i w_j = 0.308642$

$$[K]_9^e=\begin{bmatrix}
0.6571 & 0.01937 & -3.033 & -0.03847 & -4.909 & -0.7462 & 5.779 & -1.208 & 1.976 & 2.413 & -0.4696 & -0.4403\\
0.01937 & 0.1888 & -0.05546 & -0.8668 & -0.5456 & -1.463 & -0.7481 & 1.538 & 1.628 & 0.776 & -0.2981 & -0.1727\\
-3.033 & -0.05546 & 14.01 & 0.02069 & 22.59 & 3.193 & -26.84 & 5.879 & -8.835 & -11.05 & 2.116 & 2.01\\
-0.03847 & -0.8668 & 0.02069 & 4.002 & 2.14 & 6.467 & 3.906 & -7.643 & -7.369 & -2.573 & 1.341 & 0.6135\\
-4.909 & -0.5456 & 22.59 & 2.14 & 37.53 & 8.544 & -41.2 & 5.44 & -18.12 & -19.14 & 4.117 & 3.558\\
-0.7462 & -1.463 & 3.193 & 6.467 & 8.544 & 14.33 & 0.24 & -5.082 & -13.87 & -17.71 & 2.639 & 3.456\\
5.779 & -0.7481 & -26.84 & 3.906 & -41.2 & 0.24 & 55.33 & -18.83 & 9.677 & 18.69 & -2.736 & -3.257\\
-1.208 & 1.538 & 5.879 & -7.643 & 5.44 & -5.082 & -18.83 & 28.21 & 10.4 & -20.47 & -1.677 & 3.445\\
1.976 & 1.628 & -8.835 & -7.369 & -18.12 & -13.87 & 9.677 & 10.4 & 19.09 & 11.59 & -3.794 & -2.376\\
2.413 & 0.776 & -11.05 & -2.573 & -19.14 & -17.71 & 18.69 & -20.47 & 11.59 & 48.98 & -2.509 & -9.001\\
-0.4696 & -0.2981 & 2.116 & 1.341 & 4.117 & 2.639 & -2.736 & -1.677 & -3.794 & -2.509 & 0.767 & 0.5051\\
-0.4403 & -0.1727 & 2.01 & 0.6135 & 3.558 & 3.456 & -3.257 & 3.445 & -2.376 & -9.001 & 0.5051 & 1.659
\end{bmatrix}$$

则$[K]^e=[K]_1^e+[K]_2^e+[K]_3^e+[K]_4^e+[K]_5^e+[K]_6^e+[K]_7^e+[K]_8^e+[K]_9^e$

$$=\begin{bmatrix}
125.6 & 46.02 & -43.51 & 25.58 & -14.68 & -1.418 & 28.14 & 5.646 & -83.92 & -42.02 & -11.62 & -33.81\\
46.02 & 128.9 & -0.0566 & 16.67 & 4.993 & -14.68 & 5.646 & 15.25 & -42.02 & -42.74 & -14.58 & -103.4\\
-43.51 & -0.0566 & 372.7 & 49.8 & 16.67 & 25.58 & -117.5 & 33.76 & -223.9 & -142.8 & -4.431 & 33.76\\
25.58 & 16.67 & 49.8 & 372.7 & -0.0566 & -43.51 & 33.76 & -4.431 & -142.8 & -223.9 & 33.76 & -117.5\\
-14.68 & 4.993 & 16.67 & -0.0566 & 128.9 & 46.02 & -103.4 & -14.58 & -42.74 & -42.02 & 15.25 & 5.646\\
-1.418 & -14.68 & 25.58 & -43.51 & 46.02 & 125.6 & -33.81 & -11.62 & -42.02 & -83.92 & 5.646 & 28.14\\
28.14 & 5.646 & -117.5 & 33.76 & -103.4 & -33.81 & 158.2 & -34.24 & 50.57 & 38.21 & -15.99 & -9.565\\
5.646 & 15.25 & 33.76 & -4.431 & -14.58 & -11.62 & -34.24 & 69.71 & 12.57 & -52.93 & -3.155 & -15.99\\
-83.92 & -42.02 & -223.9 & -142.8 & -42.74 & -42.02 & 50.57 & 12.57 & 352.9 & 176.1 & -52.93 & 38.21\\
-42.02 & -42.74 & -142.8 & -223.9 & -42.02 & -83.92 & 38.21 & -52.93 & 176.1 & 352.9 & 12.57 & 50.57\\
-11.62 & -14.58 & -4.431 & 33.76 & 15.25 & 5.646 & -15.99 & -3.155 & -52.93 & 12.57 & 69.71 & -34.24\\
-33.81 & -103.4 & 33.76 & -117.5 & 5.646 & 28.14 & -9.565 & -15.99 & 38.21 & 50.57 & -34.24 & 158.2
\end{bmatrix}$$

位移约束： $u_1=v_3=v_4=u_6=0$

等效结点载荷：

在 1-2-3 边界上， $s=a,\ t=-1$

则： $[N_c]=\begin{bmatrix}\frac{a(a-1)}{2} & 0 & 1-a^2 & 0 & \frac{a(a+1)}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\ 0 & \frac{a(a-1)}{2} & 0 & 1-a^2 & 0 & \frac{a(a+1)}{2} & 0 & 0 & 0 & 0 & 0 & 0\end{bmatrix}$

$$x=\frac{3}{2}\sqrt{2}(a-1)a(+1)\quad \frac{3}{2}a\ a(+1)\quad y=\frac{3}{2}a\ (-a1)\quad \sqrt{\frac{3}{2}}2\ a\ (-$$

$$\therefore |J_c|=\sqrt{(\frac{dx}{da})^2+(\frac{dy}{da})^2}=\sqrt{420.4a^2+612.5}$$

在 1-2-3 边界上， $\begin{Bmatrix}q_n\\q_t\end{Bmatrix}=\begin{Bmatrix}-0.001\\0\end{Bmatrix}$， 有 $\begin{Bmatrix}q_x\\q_y\end{Bmatrix}=\begin{bmatrix}n_x & -n_y\\ n_y & n_x\end{bmatrix}\begin{Bmatrix}q_n\\q_t\end{Bmatrix}$

其中， $n_x=\frac{dy}{da}/J_c$, $n_y=-\frac{dx}{da}/J_c$

等效结点载荷：$\{r_q\}=\int[N_c]^T\begin{Bmatrix}q_x\\q_y\end{Bmatrix}dc=\int_{-1}^{1}[N_c]^T\begin{Bmatrix}q_x\\q_y\end{Bmatrix}|J_c|da$

记 $g(a)=[N_c]^T\begin{Bmatrix}q_x\\q_y\end{Bmatrix}|J_c|$

采用 3 点高斯积分，

则 $\{r_q\}=\sum_{i=1}^{3}w_i g(a_i)$

积分点 1：$a_i=-0.774597$，$w_i=0.555556$

积分点 2：$a_i=0$，$w_i=0.888889$

积分点 3：$a_i=0.774597$，$w_i=0.555556$

由此计算得 $\{r_q\}=(0,0.01067,0.02333,0.02333,0.01067,0,0,0,0,0,0,0)^T$

则单元刚度方程为：

$$\begin{bmatrix}
125.6 & 46.02 & -43.51 & 25.58 & -14.68 & -1.418 & 28.14 & 5.646 & -83.92 & -42.02 & -11.62 & -33.81\\
46.02 & 128.9 & -0.0566 & 16.67 & 4.993 & -14.68 & 5.646 & 15.25 & -42.02 & -42.74 & -14.58 & -103.4\\
-43.51 & -0.0566 & 372.7 & 49.8 & 16.67 & 25.58 & -117.5 & 33.76 & -223.9 & -142.8 & -4.431 & 33.76\\
25.58 & 16.67 & 49.8 & 372.7 & -0.0566 & -43.51 & 33.76 & -4.431 & -142.8 & -223.9 & 33.76 & -117.5\\
-14.68 & 4.993 & 16.67 & -0.0566 & 128.9 & 46.02 & -103.4 & -14.58 & -42.74 & -42.02 & 15.25 & 5.646\\
-1.418 & -14.68 & 25.58 & -43.51 & 46.02 & 125.6 & -33.81 & -11.62 & -42.02 & -83.92 & 5.646 & 28.14\\
28.14 & 5.646 & -117.5 & 33.76 & -103.4 & -33.81 & 158.2 & -34.24 & 50.57 & 38.21 & -15.99 & -9.565\\
5.646 & 15.25 & 33.76 & -4.431 & -14.58 & -11.62 & -34.24 & 69.71 & 12.57 & -52.93 & -3.155 & -15.99\\
-83.92 & -42.02 & -223.9 & -142.8 & -42.74 & -42.02 & 50.57 & 12.57 & 352.9 & 176.1 & -52.93 & 38.21\\
-42.02 & -42.74 & -142.8 & -223.9 & -42.02 & -83.92 & 38.21 & -52.93 & 176.1 & 352.9 & 12.57 & 50.57\\
-11.62 & -14.58 & -4.431 & 33.76 & 15.25 & 5.646 & -15.99 & -3.155 & -52.93 & 12.57 & 69.71 & -34.24\\
-33.81 & -103.4 & 33.76 & -117.5 & 5.646 & 28.14 & -9.565 & -15.99 & 38.21 & 50.57 & -34.24 & 158.2
\end{bmatrix}\begin{Bmatrix}0\\v_1\\u_2\\v_2\\u_3\\0\\u_4\\0\\u_5\\v_5\\0\\v_6\end{Bmatrix}=\begin{Bmatrix}f_{1x}\\0.01067\\0.02333\\0.02333\\0.01067\\f_{3y}\\0\\f_{4y}\\0\\0\\f_{6x}\\0\end{Bmatrix}$$

划去 0 位移所对应的行和列，得到降阶方程：

$$\begin{bmatrix}
128.9 & -0.0566 & 16.67 & 4.993 & 5.646 & -42.02 & -42.74 & -103.4\\
-0.0566 & 372.7 & 49.8 & 16.67 & -117.5 & -223.9 & -142.8 & 33.76\\
16.67 & 49.8 & 372.7 & -0.0566 & 33.76 & -142.8 & -223.9 & -117.5\\
4.993 & 16.67 & -0.0566 & 128.9 & -103.4 & -42.74 & -42.02 & 5.646\\
5.646 & -117.5 & 33.76 & -103.4 & 158.2 & 50.57 & 38.21 & -9.565\\
-42.02 & -223.9 & -142.8 & -42.74 & 50.57 & 352.9 & 176.1 & 38.21\\
-42.74 & -142.8 & -223.9 & -42.02 & 38.21 & 176.1 & 352.9 & 50.57\\
-103.4 & 33.76 & -117.5 & 5.646 & -9.565 & 38.21 & 50.57 & 158.2
\end{bmatrix}\begin{Bmatrix}v_1\\u_2\\v_2\\u_3\\u_4\\u_5\\v_5\\v_6\end{Bmatrix}=\begin{Bmatrix}0.01067\\0.02333\\0.02333\\0.01067\\0\\0\\0\\0\end{Bmatrix}$$

解得：$v_1=0.0003793$, $u_2=0.0002709$, $v_2=0.0002709$, $u_3=0.0003793$

$u_4=0.0002819$, $u_5=0.0002013$, $v_5=0.0002013$, $v_6=0.0002819$

2）由$u=\sum_{i=1}^{6}N_iu_i \qquad v=\sum_{i=1}^{6}N_iv_i$

在高斯积分点（-0.774597，-0.774597）处，结点位移$u=0.0000731,\ v=0.0003584$

3）在高斯积分点（-0.774597，-0.774597）处，

由$\{\sigma\}=[D]\{\varepsilon\}=[D][B]\{\delta\}^e$

计算得应力$\begin{Bmatrix}\sigma_x\\ \sigma_y\\ \tau_{xy}\end{Bmatrix}=\begin{Bmatrix}0.0024404\\ 0.0001169\\ -0.0004964\end{Bmatrix}$

$\sigma_z=\mu(\sigma_x+\sigma_y)=0.0007672$

应力矩阵$[S]=\begin{bmatrix}0.0024404 & -0.0004964 & 0\\ -0.0004964 & 0.0001169 & 0\\ 0 & 0 & 0.0007672\end{bmatrix}$

求其特征值，得到主应力：σ_1=0.0025420，σ_2=0.0007672，σ_3=0.0000153

Mises 等效应力：$\sigma_e=\frac{1}{\sqrt{2}}\sqrt{(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2}=0.0022472$